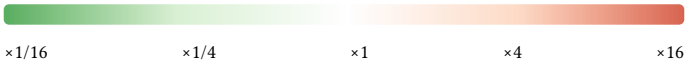
# pyAmpliCol at one loop

Batched Born–loop interference · implementation & live benchmarks

2026-10-19 · timings 2026-10-18 · AMD EPYC 9754

**One grid, four providers.** Each coloured box gives **µs per point** at $B = 1$ or $B = 128$, plus its multiplier against the fastest existing matched reference (ML, Recola or OL). Neutral lines show generation/setup and separately clocked stages at the indicated batch size. Binary64 results include Born, all helicities, full SU(3), UV, model-rule $R_2$ and all three Laurent coefficients.

| Process | pyAmpliCol | MadLoop | Recola | OpenLoops |
|---|---|---|---|---|
| $u\bar{u} \to e^-e^+$<br>photon only<br>1 loop<br>1 family | $B = 1$ **583.7**<br>**×12.46** vs ML · 46.86 µs<br>$B = 128$ **147.4**<br>**×3.17** vs ML · 46.49 µs<br>**Generation 2.43 s**<br>source + complete plan<br>first $B = 3$: 0.022 s<br>peak RSS 132 MiB<br>**Stages at $B = 32$ · µs/point**<br>Numerator 24.67<br>NinjaRS only 0.92<br>Scalar masters 0.35<br>Other work 71.69<br>Clocked total 96.84 | $B = 1$ **46.86**<br>**×1.00** vs ML · 46.86 µs<br>$B = 128$ **46.49**<br>**×1.00** vs ML · 46.49 µs<br>**Generation 198.14 s**<br>export + static compilation<br>includes bundled reducers<br>**Stages at $B = 32$ · µs/point**<br>Numerator 6.47<br>Reduction + masters 35.86<br>Shared HELAS is outside the numerator clock. | $B = 1$ **1878**<br>**×40.07** vs ML · 46.86 µs<br>$B = 128$ **1784**<br>**×38.37** vs ML · 46.49 µs<br>**Generation 0.135 s**<br>parameters + define + generate | $B = 1$ —<br>Timing not recorded<br>$B = 128$ —<br>Timing not recorded<br>**Setup —**<br>matched control unavailable |
| $u\bar{u} \to d\bar{d}$<br>14 loops<br>8 families | $B = 1$ **1733**<br>**×6.90** vs ML · 251.3 µs<br>$B = 128$ **764.0**<br>**×3.00** vs ML · 254.3 µs<br>**Generation 7.09 s**<br>source + complete plan<br>first $B = 3$: 0.063 s<br>peak RSS 180 MiB<br>**Stages at $B = 32$ · µs/point**<br>Numerator 294.0<br>NinjaRS only 7.76<br>Scalar masters 9.05<br>Other work 381.0<br>Clocked total 691.4 | $B = 1$ **251.3**<br>**×1.00** vs ML · 251.3 µs<br>$B = 128$ **254.3**<br>**×1.00** vs ML · 254.3 µs<br>**Generation 14.43 s**<br>export + static compilation<br>**Stages at $B = 32$ · µs/point**<br>Numerator 55.76<br>Reduction + masters 197.1<br>Shared HELAS is outside the numerator clock. | $B = 1$ **2651**<br>**×10.55** vs ML · 251.3 µs<br>$B = 128$ **2672**<br>**×10.51** vs ML · 254.3 µs<br>**Generation 0.253 s**<br>parameters + define + generate | $B = 1$ **547.7**<br>**×2.18** vs ML · 251.3 µs<br>$B = 128$ **563.3**<br>**×2.22** vs ML · 254.3 µs<br>**Setup 0.038 s**<br>prebuilt library · register + start<br>source generation not timed |
| $u\bar{u} \to t\bar{t}$<br>14 loops<br>9 families | $B = 1$ **2006**<br>**×5.70** vs ML · 352.0 µs<br>$B = 128$ **745.0**<br>**×1.98** vs ML · 376.2 µs<br>**Generation 6.39 s**<br>source + complete plan<br>first $B = 3$: 0.091 s<br>peak RSS 183 MiB<br>**Stages at $B = 32$ · µs/point**<br>Numerator 334.9<br>NinjaRS only 16.10<br>Scalar masters 18.28<br>Other work 520.8<br>Clocked total 889.8 | $B = 1$ **352.0**<br>**×1.00** vs ML · 352.0 µs<br>$B = 128$ **376.2**<br>**×1.00** vs ML · 376.2 µs<br>**Generation 17.05 s**<br>export + static compilation<br>**Stages at $B = 32$ · µs/point**<br>Numerator 94.48<br>Reduction + masters 256.9<br>Shared HELAS is outside the numerator clock. | $B = 1$ **2908**<br>**×8.26** vs ML · 352.0 µs<br>$B = 128$ **3053**<br>**×8.12** vs ML · 376.2 µs<br>**Generation 1.122 s**<br>parameters + define + generate | $B = 1$ **654.6**<br>**×1.86** vs ML · 352.0 µs<br>$B = 128$ **686.4**<br>**×1.82** vs ML · 376.2 µs<br>**Setup 0.010 s**<br>prebuilt library · register + start<br>source generation not timed |




Logarithmic scale · lower is faster · white is parity · red is slower

MEASUREMENTS · SAME GRID, CONTINUED

# Complete evaluation · remaining processes

Warm boxes: µs per point at $B = 1$ and $B = 128$. Neutral preparation: seconds. Instrumented stage lines: $B = 32$ for two-to-two, $B = 128$ for five-leg; no stage speed ratios.

| Process | pyAmpliCol | MadLoop | Recola | OpenLoops |
|---|---|---|---|---|
| $u\bar{u} \to gg$<br>45 loops<br>20 families | $B = 1$ **6835**<br>**×14.77** vs ML · 462.8 µs<br>$B = 128$ **3100**<br>**×8.34** vs ML · 371.5 µs<br>**Generation 21.16 s**<br>source + complete plan<br>first $B = 3$: 0.616 s<br>peak RSS 321 MiB<br>**Stages at $B = 32$ · µs/point**<br>Numerator 1778<br>NinjaRS only 21.52<br>Scalar masters 17.65<br>Other work 1389<br>Clocked total 3196 | $B = 1$ **462.8**<br>**×1.00** vs ML · 462.8 µs<br>$B = 128$ **371.5**<br>**×1.00** vs ML · 371.5 µs<br>**Generation 29.11 s**<br>export + static compilation<br>**Stages at $B = 32$ · µs/point**<br>Numerator 205.5<br>Reduction + masters 127.3<br>Shared HELAS is outside the numerator clock. | $B = 1$ **5343**<br>**×11.55** vs ML · 462.8 µs<br>$B = 128$ **4470**<br>**×12.03** vs ML · 371.5 µs<br>**Generation 0.183 s**<br>parameters + define + generate | $B = 1$ **1702**<br>**×3.68** vs ML · 462.8 µs<br>$B = 128$ **1696**<br>**×4.56** vs ML · 371.5 µs<br>**Setup 0.015 s**<br>prebuilt library · register + start<br>source generation not timed |
| $gg \to gg$<br>186 loops<br>52 families | $B = 1$ **33 656**<br>**×12.98** vs ML · 2593 µs<br>$B = 128$ **14 280**<br>**×6.89** vs ML · 2073 µs<br>**Generation 98.74 s**<br>source + complete plan<br>first $B = 3$: 11.532 s<br>peak RSS 1085 MiB<br>**Stages at $B = 32$ · µs/point**<br>Numerator 9185<br>NinjaRS only 88.42<br>Scalar masters 68.76<br>Other work 4515<br>Clocked total 14 167 | $B = 1$ **2593**<br>**×1.00** vs ML · 2593 µs<br>$B = 128$ **2073**<br>**×1.00** vs ML · 2073 µs<br>**Generation 30.14 s**<br>export + static compilation<br>**Stages at $B = 32$ · µs/point**<br>Numerator 1916<br>Reduction + masters 824.8<br>Shared HELAS is outside the numerator clock. | $B = 1$ **36 056**<br>**×13.91** vs ML · 2593 µs<br>$B = 128$ **33 245**<br>**×16.03** vs ML · 2073 µs<br>**Generation 0.170 s**<br>parameters + define + generate | $B = 1$ **11 875**<br>**×4.58** vs ML · 2593 µs<br>$B = 128$ **16 233**<br>**×7.83** vs ML · 2073 µs<br>**Setup 0.013 s**<br>prebuilt library · register + start<br>source generation not timed |
| $u\bar{u} \to ggg$<br>556 loops<br>119 families | $B = 1$ **194 468**<br>**×5.39** vs ML · 36 056 µs<br>$B = 128$ **125 357**<br>**×3.74** vs ML · 33 494 µs<br>**Generation 862.87 s**<br>source + complete plan<br>first $B = 3$: 18.424 s<br>peak RSS 5329 MiB<br>**Stages at $B = 128$ · µs/point**<br>Numerator 55 494<br>NinjaRS only 268.1<br>Scalar masters 288.4<br>Other work 73 757<br>Clocked total 129 626 | $B = 1$ **36 056**<br>**×1.00** vs ML · 36 056 µs<br>$B = 128$ **33 494**<br>**×1.00** vs ML · 33 494 µs<br>**Generation 23.89 s**<br>export + static compilation<br>**Stages at $B = 128$ · µs/point**<br>Numerator 26 380<br>Reduction + masters 2466<br>Shared HELAS is outside the numerator clock. | $B = 1$ **81 544**<br>**×2.26** vs ML · 36 056 µs<br>$B = 128$ **88 558**<br>**×2.64** vs ML · 33 494 µs<br>**Generation 0.724 s**<br>parameters + define + generate | $B = 1$ **36 928**<br>**×1.02** vs ML · 36 056 µs<br>$B = 128$ **43 380**<br>**×1.30** vs ML · 33 494 µs<br>**Setup 0.011 s**<br>prebuilt library · register + start<br>source generation not timed |

Original five processes retain revision fa599885 the five-leg row uses f89249d2. These are complete model-rule development observations; see the continued grid and measurement boundaries on the next page. The new shared-weight feature snapshot has not been retimed; these cells retain their original schedules.

MEASUREMENTS · SAME GRID, CONTINUED

# Real-mass electroweak control

Full Gmu electroweak correction, real masses, source compiler revision 85fb0900. Warm boxes: µs per point. Generation: seconds. Separate stage clocks: $B = 128$.

| Process | pyAmpliCol | MadLoop | Recola | OpenLoops |
|---|---|---|---|---|
| $e^-e^+ \to \mu^-\mu^+$<br>full EW · Gmu<br>98 loops<br>24 families | $B = 1$ **6589**<br>**×2.12** vs Recola · 3105 µs<br>$B = 128$ **2922**<br>**×0.99** vs Recola · 2946 µs<br>**Generation 115.82 s**<br>source + complete plan<br>first $B = 3$: 0.179 s<br>peak RSS 711 MiB<br>**Stages at $B = 128$ · µs/point**<br>Numerator 1539<br>NinjaRS only 31.27<br>Scalar masters 67.02<br>Other work 1284<br>Clocked total 2926 | $B = 1$ —<br>Timing not recorded<br>$B = 128$ —<br>Timing not recorded<br>**Generation —**<br>not yet recorded<br>**Stages at $B = 128$ · µs/point**<br>Numerator —<br>Reduction + masters —<br>Shared HELAS is outside the numerator clock. | $B = 1$ **3105**<br>**×1.00** vs Recola · 3105 µs<br>$B = 128$ **2946**<br>**×1.00** vs Recola · 2946 µs<br>**Generation 0.175 s**<br>parameters + define + generate | $B = 1$ —<br>Timing not recorded<br>$B = 128$ —<br>Timing not recorded<br>**Setup —**<br>matched control unavailable |

The matched Recola setup retains actual poles and the same Gmu inputs. MadLoop validates complete components; its timing and OpenLoops timing are not recorded here. No provider speed is inferred from correctness tests.

The EW row uses $G_\mu = 1.16639 \times 10^{-5}$ GeV$^{-2}$, $m_W = 80.419$, $m_Z = 91.188$, $m_H = 125$, $m_t = 173.3$ GeV, zero widths and light masses, and $\mu_R = 1000$ GeV. Its full source counterterms are retained.

## Numerator, reduction and masters

pyAmpliCol numerator includes loop DAGs, callbacks and coefficient-head preparation. NinjaRS excludes callbacks and masters; other work includes states, Born, UV/$R_2$, exact kinematics, assembly and transport. ML excludes shared HELAS from numerator and includes interfaces/retries in reduction + masters.

## Separate clocks, separate observations

Warm totals use uninstrumented calls. Stage medians come from separate clocked calls and need not sum to the median clocked total. Master clocks retain the adapter cache policy. Rust dev: local opt3 / dependencies opt2, checks on, no LTO; Community opt2. Raw timing samples are retained.

Five warm calls on repeated three-point streams; no cached event results in pyAmpliCol. CPU 382, one thread/shared host. pyAmpliCol includes decimal ingress/ Python scheduling with rescue off; providers retain native-array input and normal rescue/cache policies. Formatting is excluded. Preparation: CPU 380, ML DY 381, Recola DY and EW 382; no cache flush or uncertainty estimate. Recola photon-only DY passes analytic checks; OL DY is absent.

**Development comparison:** the measured QCD rows remain slower than ML. Acceptance requires distinct points, matched boundaries/stability, faster than ML and within 1.30 × Recola/OL per designated high-multiplicity case; generation has no gate. OL actual-pole mode runs the amplitude three times with Collier cache off. Original loop kernels use `-O0`; a separate `-O3` gg trial showed no clear win.

MEASUREMENTS · SAME GRID, NEXT VALIDATED ROWS

# New physics controls · timings pending

**New complete controls.** Six-leg processes and mixed-Born four-quark scattering pass double, DoubleFloat and arbitrary precision with effective-rule $R_2$ and UV. The same provider grid records their validation status below. Runtime, generation and stage measurements remain pending; instrumented correctness durations are not benchmark observations.

| Process | pyAmpliCol | MadLoop | Recola | OpenLoops |
|---|---|---|---|---|
| $u\bar{u} \to e^- e^+ gg$<br>full $\gamma/Z$<br>528 loops<br>71 families<br>up to 5 propagators | **Validated**<br>**double / DD / AP80**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator, NinjaRS and masters: pending | **Matched**<br>**Born and poles**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator and reduction + masters: pending | **Matched**<br>**Born and poles**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending | **Not yet matched**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending |
| $u\bar{u} \to d\bar{d}s\bar{s}$<br>pure QCD<br>275 loops<br>94 families<br>8 six-propagator families | **Validated**<br>**double / DD / AP80**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator, NinjaRS and masters: pending | **Matched**<br>**Ninja and Collier**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator and reduction + masters: pending | **Matched**<br>**cache checked**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending | **Not yet matched**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending |
| $u\bar{u} \to d\bar{d}$<br>QCD correction<br>full CKM / mixed Born<br>3 Born + 3 virtual orders<br>new physics scope;<br>prior timings unchanged | **Validated**<br>**double / DD / AP70**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator, NinjaRS and masters: pending | **Matched**<br>**every order / pole**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending<br>Numerator and reduction + masters: pending | **Not yet matched**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending | **Not yet matched**<br>Runtime $B = 1, 128$<br>Pending<br>Generation / setup<br>Pending |

## Independent physics and precision evidence

Three exact on-shell points/scales test complete and separate bare/UV/$R_2$/field terms. Six-leg DD pole/scaling controls pass $3 \times 10^{-25}$; AP80 controls pass $10^{-60}$. Six-propagator DD/AP80 outputs converge within $6.99 \times 10^{-29}$ relative. Providers remain binary64 references. Every reduction channel, empty batch and lane permutation is checked. Structural mass-shell bindings now also make the original rounded dilepton-plus-two-gluon points pass all carriers. Mixed-Born QCD checks match MadLoop within $6.7 \times 10^{-15}$, including charged chiral $R_2$ and every squared order; inclusive/resolved totals and sub-binary64 coupling updates pass all carriers.

## Provider cache and exceptional Gram configurations

Starting Recola on a special symmetric six-fermion point can affect later cached values; asymmetric priming restores uncached agreement. Recola already resets Collier per event; equivalence learning is an inferred mechanism, not an isolated upstream defect. At that Gram control, MadLoop/Ninja disagrees with Collier/Recola despite a small accuracy estimate; pyAmpliCol/NinjaRS matches the latter within $7.71 \times 10^{-14}$. Generic controls match all four runs within $1.53 \times 10^{-11}$. Failing results remain recorded; no evidence invalidates previous timing rows, and no provider source or measured observation changes.

MEASUREMENTS · MATCHED LEADING COLOUR

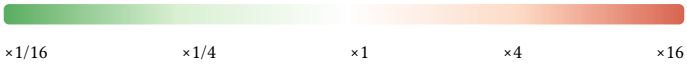
# Leading colour and its full-colour counterpart

**An internal comparison.** Each warm cell pairs LC with a new full-colour measurement of the same process and $R_2$ strategy. Its colour and multiplier refer only to that full-colour run; no external provider enters this page. Both binary64 paths return Born and all three completed Laurent coefficients.

| Process / strategy | Single point · $B = 1$ | Native batch · $B = 128$ | Generation · seconds |
|---|---|---|---|
| $u\bar{u} \to e^- e^+$<br>photon only<br>Model-rule $R_2$ | LEADING COLOUR<br>359.6 µs/point<br>Full: **351.2** µs/point<br>LC / full: ×**1.02** | LEADING COLOUR<br>103.4 µs/point<br>Full: **98.13** µs/point<br>LC / full: ×**1.05** | LC: **1.61** s<br>Full: **1.61** s<br>First $B = 3$, LC / full: 0.009 / 0.010 s |
| $u\bar{u} \to d\bar{d}$<br>Model-rule $R_2$ | LEADING COLOUR<br>1635 µs/point<br>Full: **1536** µs/point<br>LC / full: ×**1.06** | LEADING COLOUR<br>609.0 µs/point<br>Full: **646.7** µs/point<br>LC / full: ×**0.94** | LC: **6.05** s<br>Full: **5.27** s<br>First $B = 3$, LC / full: 0.058 / 0.057 s |

**Stages · LC / full · µs/point**

| | $u\bar{u} \to e^- e^+$ | $u\bar{u} \to d\bar{d}$ |
|---|---|---|
| Numerator | 25.94 / 23.11 | 287.4 / 303.1 |
| NinjaRS only | 0.66 / 0.64 | 6.30 / 6.17 |
| Masters | 0.20 / 0.21 | 6.83 / 6.53 |
| Other | 76.81 / 70.79 | 355.4 / 368.5 |
| Clocked total | 103.8 / 94.67 | 658.3 / 693.9 |




Logarithmic scale · lower is faster · white is parity · red is slower

Independent fresh process for each colour mode, revision 95c4ed50, CPU 379, one thread/shared host. Five timed calls use the repeated three-point controls; decimal ingress and Python batch scheduling are included, checks and output formatting excluded. Generation includes source capture and the complete plan; first evaluation is separate. These are development observations, not an LC performance target or a replacement for the full-colour snapshot on pages 1–3.

Fixed-$N_f$ LC keeps the common highest formal-$N_c$ power before setting $N_c = 3$. Full and LC retain the current graph schedules; this is not yet numerator pruning. Quark controls use the fixed-SU(3) full path and formal LC path. The gluon control uses the certified direct strategy and formal algebra for both paths. New source model-rule gluon checks pass full/LC/NLC in all carriers, with timings pending; they do not relabel these observations. Full components pass independent references; LC checks use the certified Born, pole and UV/field relations, with complete component reconstruction.

MEASUREMENTS · SAME COLOUR GRID, CONTINUED

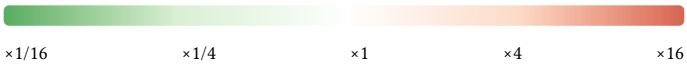
# Leading colour · massive quarks and gluons

Warm cells: µs per point. Neutral B128 stage pairs: LC / full, from separate instrumented calls; medians need not add. Generation and first evaluation: seconds.

| Process / strategy | Single point · $B = 1$ | Native batch · $B = 128$ | Generation · seconds |
|---|---|---|---|
| $u\bar{u} \to t\bar{t}$<br>Model-rule $R_2$ | LEADING COLOUR<br>1729 µs/point<br>Full: **1941** µs/point<br>LC / full: ×**0.89** | LEADING COLOUR<br>723.7 µs/point<br>Full: **730.0** µs/point<br>LC / full: ×**0.99** | LC: **5.64** s<br>Full: **5.25** s<br>First $B = 3$, LC / full: 0.078 / 0.084 s |
| $gg \to gg$<br>Direct $R_2$ | LEADING COLOUR<br>34 688 µs/point<br>Full: **34 287** µs/point<br>LC / full: ×**1.01** | LEADING COLOUR<br>14 570 µs/point<br>Full: **18 843** µs/point<br>LC / full: ×**0.77** | LC: **126.59** s<br>Full: **161.07** s<br>First $B = 3$, LC / full: 13.941 / 13.612 s |

**Stages · LC / full · µs/point**

| | $u\bar{u} \to t\bar{t}$ | $gg \to gg$ |
|---|---|---|
| Numerator | 291.5 / 293.5 | 12 005 / 11 703 |
| NinjaRS only | 10.00 / 10.01 | 161.2 / 144.0 |
| Masters | 13.06 / 12.46 | 110.4 / 104.3 |
| Other | 433.1 / 437.9 | 2895 / 4050 |
| Clocked total | 747.9 / 752.4 | 15 302 / 16 133 |



×1/16 ×1/4 ×1 ×4 ×16

Logarithmic scale · lower is faster · white is parity · red is slower

**What is and is not measured.** Every timed call computes the complete selected observable. LC changes the retained formal-colour powers; current graph schedules still run. These same-strategy internal pairs do not compare direct and model $R_2$, nor replace the model-rule provider grid. Stage scopes match that grid: numerator includes DAGs and reducer callbacks/head preparation; NinjaRS excludes callbacks and scalar masters. Other work includes states, Born weighting, counterterms, kinematics, assembly and instrumentation.

**Opt-in projector JIT.** Saved family bilinears now bind through the existing tiled coefficient plan when `projector_engine="jit"` is requested in binary64. The interpreter remains the default and the DoubleFloat/AP path. A fresh complete gluon check at `d6c05e08`, on 128 distinct exact energy-scaled control points, gave $B = 128$ projection medians of 1.75 to 0.52 ms/point and complete calls of 13.28 to 11.85 ms/point. Five paired CPU379 blocks retain visible variation; independent physics gates passed unchanged. Saved ordering, retained/portable bindings, tail/empty batches and precision checks pass. JIT is not bit-exact; earlier output-wise DD evidence remains available. No algebra, arena or factorization change is included. These observations do not replace the timing tables.

ARCHITECTURE · PHYSICAL CONTRACT

# The quantity being computed

The new engine targets one-loop amplitudes interfered with a Born amplitude. It does not compute the squared one-loop amplitude and does not admit loop-induced production. A Born zero at an individual point or helicity remains a valid event of a supported process.

For Born order $a$, loop order $b$ and squared coupling order $s = a + b$, the ordinary uncorrelated observable is

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\Re\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0, -1, -2\}.$$

The rectangular matrix $G_{BL}$ connects the actual Born and loop colour bases. The normalization $\mathcal{N}$ includes declared initial averages and final identical-particle factors. Couplings are included; the result is neither divided by the Born value nor normalized by an extracted strong coupling.

The returned coefficient order is **(finite, single pole, double pole)** in the MadLoop convention,

$$V = C_\varepsilon \left( V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2} \right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}.$$

The finite conversion used in the matched Recola controls is $V_0^{\mathrm{ML}} = V_0^{\mathrm{Recola}} + \left(\frac{\pi^2}{6}\right) V_{-2}$; both poles are unchanged. Such conversions act on the Laurent series, not on a process-dependent fitted factor. Scheme, scale and finite renormalization conventions are retained separately.

## Coupling selection is a graph property

`loop_coupling_orders` requests perturbative corrections. Eligibility depends on the actual loop interactions and internal species, followed by explicit Born–loop pairings. An order difference alone cannot identify the correct graphs. In the mixed four-quark control,

$$B = B_{20} + B_{02}, \quad L = L_{40} + L_{22} + L_{04},$$

the squared order $(4, 2)$ receives both $B_{20}^* L_{22}$ and $B_{02}^* L_{40}$. Their identities survive through counterterm matching and output assembly. Joint QCD/QED selection also admits mixed graphs absent from either singleton selection. Independent MadLoop graph controls give 16 QCD, 15 QED and 35 joint loops for the retained mixed model.

## What the measured processes include

| Control | Initial / identical normalization | Physical completion |
|---|---|---|
| $u\bar{u} \to e^- e^+$ | $\frac{1}{36}$ | Photon Born; QCD virtual |
| $u\bar{u} \to d\bar{d}$ | $\frac{1}{36}$ | Local UV and $R_2$ |
| $u\bar{u} \to t\bar{t}$ | $\frac{1}{36}$ | Local and external-field UV; $R_2$ |
| $u\bar{u} \to gg$ | $\frac{1}{72}$ | Local/field UV; $R_2$ |
| $gg \to gg$ | $\frac{1}{512}$ | Local/field UV; $R_2$ |
| $u\bar{u} \to ggg$ | $\frac{1}{216}$ | Local/field UV; $R_2$ |
| $e^- e^+ \to \mu^- \mu^+$ | $\frac{1}{4}$ | Full EW; real-mass Gmu; local UV/ $R_2$ |

The QCD scattering controls use $\alpha_s = 0.118$, $\mu_R = 1000$ GeV, stable $m_t = 173$ GeV, five renormalization flavours and six quarks in loops; the other quark masses vanish under explicit control restrictions. Drell–Yan uses $\alpha^{-1} = 132.507$. Initial spin and colour are averaged, and physical external helicities are summed. Two final gluons include $\frac{1}{2}$, three include $\frac{1}{6}$.

ARCHITECTURE · MODEL AND COUNTERTERMS

# Preserve the model before lowering it

Loop models contain information that a tree-only loader can discard: Laurent-valued parameters and couplings, UV and $R_2$ categories, loop-particle groups, external-field factors and duplicate vertex names. The upstream loader's raw source model retains these fields and positional identities in ordinary model JSON. pyAmpliCol consumes it through a thin adapter, with one stored source copy.

**Raw source**
Exact declarations
Species and loop groups

→

**Prepared model**
Scalar programs
Colour × Lorentz entries

→

**Process plan**
Admitted graphs
Matched UV / $R_2$ / fields

## Exactness begins at input

Literal decimal and Laurent source text, exact overrides and fixed parameters retain their scalar owner. Imported results of arbitrary Python computation or mutation do not acquire exactness. Symbolica performs substitution, series and expression preparation. Source helpers retain their branches; native evaluation checks each value and active diagnostic. A nonfinite or unsupported branch cannot silently succeed. Per-point scales reach the same counterterm scalar program.

Lowering omits a basis only when exact coupling bounds prove it irrelevant; source, restrictions and species eligibility remain retained. The explicit `loop.model` contract declares joint correction coverage, UV/$R_2$ completeness, dimensional regularization, external states, gamma-five, renormalization, masses and finite restoration. It remains `declared-not-certified`: model names and counterterm presence do not prove completeness. Public generation now consumes this explicit prepared-model contract.

## Tensor structure and physical frame

Each basis entry retains its scalar coefficient, native Lorentz tensor, native colour tensor and ordered particle ports. Multi-colour quartic vertices remain sums of local terms. There is no reconstruction of tensor algebra in pyAmpliCol: Idenso/Spenso/FeynKit own contractions, traces, component libraries and index transformations.

Source spinor slots require an explicit map to native wavefunctions. The FFV map renames typed external Dirac ports simultaneously; the FFS map also exchanges `ProjP`/`ProjM`, with 580 independent ALOHA controls. Neither rewrites momentum labels or supplies a universal gamma-five transpose. Unvalidated structures remain unsupported.

## Counterterms follow admitted loops

UV-tree terms, local loop counterterms and external-field factors have different eligibility rules. A standalone counterterm tree is insufficient evidence that its loop correction belongs to the request. Local $R_2$/UV matching retains the actual attachment currents, ordered binding legs, local coupling order and allowed loop-species alternatives. Wildcards and repeated alternatives preserve their source semantics and multiplicity. Fresh native tree graphs provide the insertion signs and propagators. Ordinary Bose slot classes are identified only after exact native tests of the complete vertex under generating transpositions. This repairs 39 missed insertions in the five-leg $u\bar{u} \to ggg$ control. Complete Born, bare, UV and $R_2$ components now agree with MadLoop at the reference accuracy in double, DoubleFloat and AP80; exact parameter-update controls also pass.

External-field factors multiply the applicable Born currents with each physical leg retained. Local and field UV remain separately inspectable, then join the same ordered interference. Model-rule $R_2$ currents already contain their loop normalization; the assembly does not multiply them by a second loop measure.

The bounded Gmu continuation offers an explicit `ufo-standard` logarithm convention. The default preserves the source helper; either choice retains the original model and does not certify complete CMS physics.

**Loader 1.0.1 is published.** Lossless source import, JSON and finite tree projection pass QCD/EW controls. `sqrt (2)` and real/imaginary parsing are repaired; SM assets need no changes. pyAmpliCol uses the released source API.

ARCHITECTURE · RECURRENCE AND FAMILIES

# Local tensors, reusable coefficient currents

The production direction is a dedicated loop recurrence. FeynKit supplies the graphs, directed incidence, loop signs, symmetry factors, cuts and affine momentum routing. pyAmpliCol turns those local ingredients into a coefficient dataflow graph and schedules its numerical evaluation.

A current retains the coefficients of a symmetric polynomial in the physical loop momentum. For a local tensor $T$ and previous current $J$, the next step is the contraction $T \cdot J$, followed by coefficient extraction using Symbolica. Only the local transition is prepared symbolically. A whole-diagram symbolic trace is used in small independent oracles, not as the numerical recurrence.

**Coefficient dataflow.** External states and tree attachments feed local transitions; loop closure produces ordered coefficient outputs. Native saved evaluator instructions update these outputs in batches. Each original graph retains its colour terms, coupling sector and routing map even when it shares an identical numerical schedule with another graph.

## Topology remains attached to the numerator

Reduction needs a denominator set as well as a polynomial. Each graph carries its routed propagators and a dependency-proved map into an integral family. Numerators are transported through that map before combination. Cyclic or reversed order is not inferred from a loose list of masses: momenta, incidence, external attachments and the actual transformation must agree.

The family plan retains bridges and local tree denominators. Native graph factors carry fermion and ghost signs. Multi-term local vertices are expanded into ordered recurrence outputs, while the original graph/family identities remain intact. This separates a numerical reuse opportunity from a physical multiplicity or symmetry factor.

## Contract before reducing when linearity permits

One exact dual-Born map now computes complex weights once per point/helicity or source row. Its columns retain direction, squared order and original loop term. Families select these columns independently of colour orbits, then assemble routed numerator coefficients before reduction. Resolved component and directed banks preserve their own metrics; UV/$R_2$/field completion is unchanged.

Complex weights are retained until the correctly directed bilinear is formed. Taking a real part too early loses information needed by colour histories and other off-diagonal observables. Counterterm interference uses the same Born currents and colour conventions, with its own local scalar factors.

## Sharing that is safe under parameter updates

Compiled kernel reuse keys the saved instructions, settings, carrier precision and output layout. Evaluators may share immutable compiled code while owning independent work buffers. Whole-schedule numerical reuse additionally requires identical ordered inputs under an exact model-binding proof. Equal floating values at one point are insufficient.

The current process cache shares compatible Born, loop and counterterm kernels; exact whole-schedule groups cover Born and loop execution. It does not cache event results between calls. Complete unprojected model-rule QCD plans now retain exact instructions and bindings across arithmetic views. Rebinding creates fresh evaluators, reducers and buffers without repeating graph, colour or coefficient preparation. Colour-weight changes invalidate the cached plan. The bounded full-colour direct-$R_2$ gluon plan also rebinds its retained dimensional DAG and physical/evanescent sectors, including UV and field completion. Source-owned complete QCD views also retain their exact scalar definitions and parameter snapshots across carrier rebinding. Saved directed projectors now use this same ownership boundary; reducer Laurent preparation remains a cold binding cost.

**Separate sharing experiment.** On distinct exact energy rescalings of gluon controls, five alternating calls at $B = 32, 128$ gave complete medians 15.03 ⟶ 19.97 / 15.55 ⟶ 21.91 ms per point. Exact interning reduced 1,963 nodes to 1,358 with bit-identical outputs, but native execution slowed. A cache-locality explanation remains an inference; global interning was not adopted. CPU 379/shared host; separate from headline data.

**Graph-local arena experiment.** Existing lifetime allocation reduced retained arena capacity by 48.4% without changing instructions or 16-row tiles. Complete $B = 32, 128$ medians were 14.23 ⟶ 13.55 / 13.60 ⟶ 14.06 ms per point. Coefficients and complete components were bit-identical; DD/AP80 layout checks passed. No speed benefit was established; the production layout is unchanged. Both plans coexisted, capacity excludes dependency buffers, and raw DAG clocks remain available. Same distinct-point protocol and CPU.

ARCHITECTURE · COLOUR AND OBSERVABLES

# Full colour without dense event tensors

Dependency algebra contracts actual Born/loop tensors into cached rectangular weights, preserving residual networks. Events evaluate these weights without dense colour tensors or numerical colour fits.

## LC and NLC are structural projections

For a certified formal-$N_c$ model, all contributions belonging to one squared order and operator family are combined before finding the common highest power $P$. With $N_f$ held fixed,

$$\mathrm{LC} : N_c^P, \quad \mathrm{NLC} : N_c^P, N_c^{P-1}, N_c^{P-2}.$$

The common $P$ spans helicities, epsilon and bare/UV/$R_2$/field terms, including odd suppressions. All carriers retain exact invariants and diagnostics; signed Born anchors reconstruct inclusive results. Fresh gg LC/NLC passes. DY has LC/full $= \frac{9}{8}$ and NLC $=$ full; DY LC uses the isolated PR 155 overlay.

An explicit general-$N_c$ replacement for `loop_sm`'s fixed-SU3 four-gluon $R_2$ matches its complete source tensor at $N_c = 3$ and independent published rules: Fig. 2 of 0903.0356 and Eqs. 5.30–5.38 of 2007.03713. Source groups, diagnostics and HV bindings survive; full/LC/NLC passes exact updates and 16-helicity sums in all carriers. Arbitrary model continuation remains uncertified.

Seven-leg exact SU(3) components close a colour network whose formal reduction retains a degree-five invariant; existing rewrites cover degrees three and four. Its trace-scope repair therefore does not certify formal colour. Upstream PR 155 separately fixes open-dual canonicalization; isolated fundamental-port controls pass.

## Directed colour and spin sources

Correlated virtual observables require both rectangular cross terms. For operator histories $A$ and $B$, their generic form is

$$B_A^\dagger G L_B + L_A^\dagger G B_B.$$

Unequal histories can give complex results, beyond $2\mathfrak{R}\left(B_A^\dagger G L_B\right)$. Existing native charge tensors act on shared currents in one batch. Four-quark controls pass all carriers, swapped terms, order/field masks and exact updates. Each direction is integrated before conjugation and verified separately, so cancellation cannot hide a bad term. Subset rescue preserves physical IDs/history labels.

Quark-scattering identity, Casimir and $T_1 \cdot T_2$ Born/Laurent values match Recola in double/DD/AP70; this is binary64 provider agreement. Complex multi-emission histories have separate controls. Model-owned photon/gluon source identities and phases survive fresh $u\bar{u} \to \gamma g$ and crossed $u\gamma \to ug$ publication. Transverse-complex singleton sources match Recola; arbitrary literal vectors use recurrence/replay controls without projection. Joint sums, exact sources, empty batches, AP60/AP80 rescue and photon SDK pass.

Projected $gg \to gg$ LC/NLC now composes with singleton/joint spin for identity histories. Every complete component and both terms reconstruct ordinary results in all carriers, retaining the common colour degree. Fresh public NLC, saved LC, public LC subset rescue and contracted-component SDK shapes pass. Ordinary LC keeps its signed Born anchors; inclusive identity correlations have one contracted component. Massive vectors and nonidentity formal-colour histories remain open.

## Exact mappings and the rectangular FFT boundary

Exact flavour reuse preserves resolved four-quark output and complex histories. Paired source/phase proofs admit $d\bar{d} \to gg$ / $s\bar{s} \to gg$ singleton/joint spin: all carriers, both terms, rescue and Rust SDK pass. LC/NLC mapping and nonidentity permutations remain open; no generation speedup is claimed.

Native FFT adjoint/inverse and cold tensor certificates cover noncentral complex $S_3$, full/LC/NLC and residual cross terms. Cold projection retains exact pre-$N_c$ rectangles, common degrees and physical tensor/order columns. Cold physical-orbit certificates and factories agree with direct ordinary/resolved weights. Shared Born weights pass all carriers, including gg LC's six signed components. Public FFT attachment, selection and execution remain open; direct weights are unretimed.

ARCHITECTURE · NATIVE BATCHES AND PRECISION

# One numerical path, three real carriers

Binary64, DoubleFloat and arbitrary precision run through coefficients, kinematics, NinjaRS, masters, counterterms and contractions. Values are never computed in double and promoted. Metadata records the actual carrier and requested precision.

Exact decimal input and rational model coefficients precede conversion. Source mass/coupling formulas and diagnostics use the chosen native carrier. Controls with changes below binary64 resolution test that information through the output. Physical source plans now bind known external mass shells structurally, using exact model mass outputs in the same carrier. This also covers the dependent external momentum; raw off-shell oracle evaluators remain unchanged.

## Native batches, including resolved helicities

Ordered layouts preserve point/helicity identities through tails, compaction and scatter. One Rust call executes the complete batch using reusable native tiles; 16 rows is a scheduling choice, not a capacity limit. Binary64 may use SymJIT; DD/AP remain native batches without an unsupported SIMD claim.

Resolved output preserves point, order, helicity, colour component and epsilon. Actual source gluons pass with 16 helicities and six signed LC components in all carriers. Currents are shared across components; only projection rows expand. `.total()` retains original averaging and every order/pole. Full/NLC keep one contracted component. Opaque IDs through the full native integer range pass Python and SDK checks. Early/late failures remove or rescue whole points. The components are signed interference terms, never sampling probabilities.

## Reduction and exact saved plans

NinjaRS receives routed denominators, polynomial coefficients, exact invariants and carrier-appropriate inputs. Weighted rank counts $\mu^2$ twice; explicit epsilon sectors combine by Laurent multiplication. The additive dense-$\mu$ adapter does not change the engine identified by each timing record.

Consumer `0d0d7e7` uses self-contained Git pins and passes saved six-propagator physics and seven-line rank-eight controls in all carriers. Its reducer owns rank admission; a saved rank-four bubble passes all carriers. Complete seven-line physics and general higher-rank models remain open. The retained producer stays on `e4c16a7`. Scalar-master PR 9 passes isolated AP80 scalar and complete zero-width CMS controls. Normal pins are unchanged; dependency adoption and full CMS scheme certification remain open.

The installed `bc78edbd` wheel passes tree, LC 16×6, DD and no-Python Rust checks. Fresh gg LC/NLC publication validates two points and reloads in AP70; DY LC needs the PR 155 Community overlay. Later rank, directed execution and ID changes have separate source/native gates; these checks add no timing measurements.

Transactional API export provides six installed mixed-order resolved clients. New source-built Rust/C/C++/Fortran and versioned UMAMI grouped clients preserve complex history/order/pole axes, literal joint spin sources, physical IDs, statuses and owned diagnostics. Source layouts, empty/partial batches and rescue pass. Fresh correlated CLI uses a matching developer build, separate from installed `bc78`.

## Successful completion is not established accuracy

Native guards retain original point IDs and exact snapshots. Public evaluation rejects unrecovered failures by default. A policy configured before binding can retain exact inputs after numerical overflow if a higher carrier succeeds. Default binding stays eager; invalid inputs and failed updates keep atomic rollback. Source admission does not establish accuracy. For rescue, two complete higher results must agree across components, orders, poles, helicities and both directed terms, followed by checked output conversion. Attempt histories survive and the $R_2$ strategy stays fixed. Guard-only controls pass native/public Python signed-LC resolved and joint-spin results, Rust rescue and C/C++/Fortran policy transport. Physical evaluators are unchanged; these checks add no physics certification or timing observations.

**Open boundaries.** Producer policy composition, CLI policy transport and explicit public partial-value return remain pending. General input-consistency diagnostics, wider physics certification and normal adoption of the tested CMS master fix remain open.

ARCHITECTURE · RATIONAL TERMS

# Two complete strategies, one observable

The model strategy combines the four-dimensional numerator contribution with the model's effective $R_2$ rules and UV terms. The direct strategy instead computes the required evanescent numerator remainder and retains the same UV completion. The strategy is explicit in the result; a caller cannot relabel an executed model result as direct.

Four-dimensional reduction already produces denominator-related rational terms. The presence of $\mu^2$ in a reducer interface therefore does not, on its own, prove that numerator-origin $R_2$ is complete. In the target HV scheme, effective rules already include the missing spin-dimension terms. They must not also be added to the four-dimensional numerator. FDH has a different prescription.

## Local evanescent state, not a whole-loop trace engine

For a certified closed vectorlike fermion cycle, the local component construction uses dependency gamma matrices with one auxiliary evanescent slash. Odd auxiliary powers survive every open transition; only the closed trace selects even parity. Independent full-$D$ local-graph oracles check routed bubbles, triangles and boxes. The auxiliary gamma-five realization is not a general physical chiral or dimensional gamma-five prescription.

For the admitted massless Yang–Mills cycles, dependency metric and momentum rules are continued to an explicit physical/aligned/spectator state space. The remaining transverse-state multiplicity is retained symbolically in $D = 4 - 2\varepsilon$. Local cubic and quartic colour terms, closure, routing and spectator weighting are checked independently. Ghost cycles keep their native statistics and their ordinary bare contribution.

## Epsilon sectors matter after integration

If the numerator is $N_0 + \varepsilon N_1 + \varepsilon^2 N_2$ and each reduced sector has finite/single/double coefficients $\left(F_j, S_j, D_j\right)$, the assembled result is

$$V_0 = F_0 + S_1 + D_2, \quad V_{-1} = S_0 + D_1, \quad V_{-2} = D_0.$$

Dropping explicit epsilon terms can therefore change the finite answer even when the mu-dependent terms are retained. Native reduction controls exercise this convolution in all three carriers. Source templates, full-graph algebra and integrated physical results have separate checks.

## What is supported internally today

The complete retained full-colour $gg \to gg$ control admits the local direct kernels, includes UV completion, and agrees with model-rule components and independent reference results in all three arithmetic modes. Bounded cross-precision verification and direct-strategy rescue are also implemented. Public direct dispatch remains behind its integration gates.

Open fermion chains, general mixed-spin loops, massive gauge/Goldstone sectors, physical axial or anomaly-sensitive structures and arbitrary BSM dimensional extensions are not inferred from that success. Unsupported direct domains need compatible model-rule completion. No universal massless FDH-to-HV shift is assumed for massive EW or general models.

**Current priority.** Complete effective-rule $R_2$ plus UV, feature coverage and higher multiplicities. Further generic direct development and strategy timing are deferred; existing bounded controls and interfaces are retained. Headline timings use model rules. The historical paired gluon LC/full cells use direct completion. Scheme distinction: QCD rational rules, §2.

VALIDATION · NEXT GATES

# What this snapshot establishes

The controls run model import, graphs, recurrence, reduction, UV/$R_2$ and ordered physical output together. Independent references separate bare, local UV, external-field UV and $R_2$, testing components before their complete sum.

| Area | Current boundary |
|---|---|
| Complete QCD | Two-to-two, five-leg gluon and both six-leg controls pass all carriers, including eight true six-propagator families. Full-CKM four-quark QCD corrections also pass all mixed Born orders with charged chiral $R_2$. New timings remain pending. |
| Real-mass EW | Source-owned neutral-current charged-lepton controls, including bounded massive muons; complete components and scale/parameter changes. |
| CMS | Gmu controls match MadLoop components across widths; isolated PR 9 now passes AP80 zero-width scalar and complete source checks. Normal adoption and wider scheme certification remain open. |
| Correlations / mapping | Directed/joint sources include crossed photons and exact d/s reuse in all carriers. Projected gg LC/NLC identity histories now compose with spin; ordered formal-colour operators, LC/NLC mappings and nonidentity permutations remain open. |
| Public delivery | Installed gg LC/NLC and AP70 reload pass; DY LC needs PR 155. Saved/public Python and standalone SDK configured source-binding rescue pass; producer composition, CLI policy and public partial returns remain open. |
| Performance | Current development timings only; distinct-point matched provider gates and required high multiplicity remain open. |

## Required multiplicity and acceptance

Selected $2 \to 5$ gluon, heavy-quark and dilepton channels remain mandatory; $2 \to 6$ is a stretch goal. Five-final-parton gluon channels require MadLoop/Recola controls; unavailable OpenLoops channels are explicitly excluded.

Full $\gamma/Z\ u\overline{u} \to e^-e^+gg$ passes all carriers at three exact on-shell points: 16 Born graphs, 528 loops, 71 families, 406 local counterterms, 96 field insertions and normalization $\frac{1}{72}$. Separate components match binary64 references; double also passes distinct RAMBO points within $6.56 \times 10^{-13}$ relative. Structural mass-shell bindings now make the original rounded points and their separate components pass DD/AP80 too.

The six-fermion control adds 7 Born graphs, 275 loops, 94 families and 341 CT matches, with normalization $\frac{1}{36}$. Its complete components pass all carriers; page 4 records precision evidence and the independent cache/Gram audit. The new timing rows remain pending. Full-CKM four-quark QCD corrections now match MadLoop for three Born and three virtual orders at three points, including the charged chiral-$R_2$ source rules. This does not certify joint QCD/EW corrections. The seven-leg control has matched independent references. Its source build completed Born and first-family projection before the retained process disappeared with no recorded terminal cause. An unchanged detached recovery is underway; no seven-leg numerical result is claimed. Required $2 \to 5$, FFT, broader spin/colour/mapping composition and general-model coverage remain open.

## Independent evidence and reproducibility

`data/measurements.json` retains the samples, revisions and cold times behind the unchanged tables; `tools/loop_reference/` contains frozen points and drivers. Timing covers complete `evaluate_ordered_native` calls, excluding checks and output conversion. New source-binding, projected-spin, photon, six-leg and shared-weight checks add no timings. The new schedule is unretimed; original timing identities are retained.

Providers are MadLoop 3.7.0, Recola2 2.3.0 with matched SM/Collier, and OpenLoops 2.1.5. Tables use matched-host control timings with input and rescue asymmetries declared. Their measured object is complete physical interference.

Further reading: repository `docs/development/LOOP_MODEL_IR.md`, `docs/development/LOOP_DIRECT_R2.md`, `docs/loop-flavor-symmetry.md`, and `tools/loop_reference/README.md`; OpenLoops on-the-fly reduction · Recola · four-dimensional formulation · MadLoop standalone convention.

**Living document.** Update source, data and PDF after each milestone. Current priority is requested feature coverage and validated two-to-four/two-to-five processes; further two-to-two optimization is deferred. Add timing rows only after physical validation. No arbitrary caps on process size or current count; report actual dependency limits explicitly.